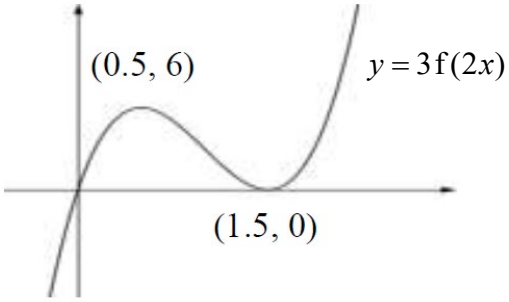
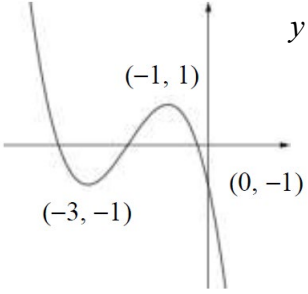
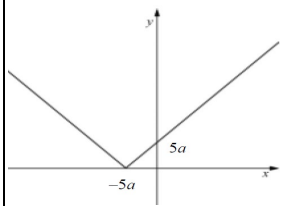
1	$\int \frac{x^2-5}{2x^3}\,\mathrm{d}x = \int Ax^{-1} - Bx^{-3}\,\mathrm{d}x = C\ln x + Dx^{-2}\,(+c)$	M1 dM1
	$= \frac{1}{2}\ln x + \frac{5}{4}x^{-2} + c$	A1
		(3)
		Total 3

2(i)		Shape (two way stretch) Maximum at (0.5, 6) Minimum at (1.5, 0)	B1 B1 B1
			(3)
(ii)		Shape and position Minimum at (−3, −1) and maximum at (−1, 1) Crosses y-axis at (0, −1)	B1 B1 B1
			(3)
			Total 6

3(a)	$2x^2-3x-5=(2x-5)(x+1)$	B1
	$3-\frac{x-2}{x+1}+\frac{5x+26}{2x^2-3x-5}=\frac{3(x+1)(2x-5)-(x-2)(2x-5)+5x+26}{(x+1)(2x-5)}$	M1 A1
	$=\frac{(4x+1)(x+1)}{(x+1)(2x-5)}=\frac{4x+1}{2x-5}$	A1
		(4)
(b)	Correct attempt at inverse $y=\frac{4x+1}{2x-5}\Rightarrow x=...$	M1
	$f^{-1}(x)=\frac{5x+1}{2x-4}$	A1
		(2)
(c)	$2<x<\frac{17}{3}$	M1 A1
		(2)
		Total 8

4(a)	Either $x = -\frac{a}{3}$ or $y = a$	B1
	Correct coordinates $\left(-\frac{a}{3}, a\right)$	B1
		(2)
(b)	 V shape in Quad 1 and 2 with the vertex on the negative x-axis	B1
	Intersects/meets axes at $(0, 5a)$ and $(-5a, 0)$	B1
		(2)
(c)	Attempts to solve a correct equation **E.g. either** $x + 5a = -3x - a + a \Rightarrow x = ...$ **or** $x + 5a = 3x + a + a \Rightarrow x = ...$	M1
	$x = -\frac{5}{4}a$ **or** $x = \frac{3}{2}a$	A1
	$x = -\frac{5}{4}a$ and $x = \frac{3}{2}a$	A1
	$x = ... \Rightarrow y = ...$ using either equation	dM1
	$\left(-\frac{5}{4}a, \frac{15}{4}a\right)$ and $\left(\frac{3}{2}a, \frac{13}{2}a\right)$	A1
		(5)
		Total 9

5(a)	$18 = A - 180 \times 1 \Rightarrow A = ...$	M1
	$A = 198$	A1
		(2)
(b)	$90 = 198 - 180\mathrm{e}^{-5k} \Rightarrow 180\mathrm{e}^{-5k} = 108$	M1, A1
	$\mathrm{e}^{-5k} = \frac{108}{180} \Rightarrow -5k = \ln 0.6 \Rightarrow k = ...$	dM1
	$k = -\frac{1}{5}\ln\frac{3}{5}$ or $k = \frac{1}{5}\ln\frac{5}{3}$	A1
		(4)
(c)	$\theta = 198 - 180\mathrm{e}^{-9\times\frac{1}{5}\ln\frac{5}{3}} \Rightarrow \theta = ...$	M1
	$\theta = 126°\mathrm{C}$ (awrt)	A1
		(2)
(d)	$\left\{\frac{\mathrm{d}\theta}{\mathrm{d}t}\right\} = -180\left(-\frac{1}{5}\ln\frac{5}{3}\right)\mathrm{e}^{-\frac{t}{5}\ln\frac{5}{3}} = ...$	B1ft
	$\left\{\frac{\mathrm{d}\theta}{\mathrm{d}t}\right\} = -180\left(-\frac{1}{5}\ln\frac{5}{3}\right)\mathrm{e}^{-\frac{9}{5}\ln\frac{5}{3}} = ...$	M1
	= 7.33 °C min^{-1} (awrt)	A 1
		(3)
		Total 11

6(a)	$\left(\mathrm{f}'(x)\right)=\cos\left(\frac{x}{3}\right)-\frac{1}{3}x\sin\left(\frac{x}{3}\right)$	M1 A1
		(2)
(b)	$\mathrm{f}'(x)=0\Rightarrow\cos\left(\frac{x}{3}\right)-\frac{1}{3}x\sin\left(\frac{x}{3}\right)=0\Rightarrow 1-\frac{1}{3}x\tan\left(\frac{x}{3}\right)=0$	M1
	$\tan\left(\frac{x}{3}\right)=\frac{3}{x}\Rightarrow\quad x=3\arctan\left(\frac{3}{x}\right)\qquad *$	A1*
		(2)
(c)	$x_2=3\arctan\left(\frac{3}{2.5}\right)=$ awrt 2.6	M1
	$x_2=$ awrt 2.628 and $x_6=$ awrt 2.586	A1
		(2)
(d)	$\mathrm{f}'(2.5815)=\cos\left(\frac{2.5815}{3}\right)-\frac{1}{3}(2.5815)\sin\left(\frac{2.5815}{3}\right)=-0.000345...$ $\mathrm{f}'(2.5805)=\cos\left(\frac{2.5805}{3}\right)-\frac{1}{3}(2.5805)\sin\left(\frac{2.5805}{3}\right)=0.000346...$ Chooses a suitable interval and attempts both values	M1
	(Both values correct) Sign change and continuous, therefore root	A1
		(2)
		Total 8

7(a)	Uses $\sin 2x = 2\sin x\cos x$ **AND** $\cos 2x = 1-2\sin^2 x$ o.e. in $\dfrac{\sin 2x}{\cos x}+\dfrac{\cos 2x}{\sin x}$	M1
	$\dfrac{\sin 2x}{\cos x}+\dfrac{\cos 2x}{\sin x}=\dfrac{2\sin x\cos x}{\cos x}+\dfrac{1-2\sin^2 x}{\sin x}$ $=\dfrac{2\sin x\cancel{\cos x}}{\cancel{\cos x}}+\dfrac{1}{\sin x}-\dfrac{2\sin\cancel{^2} x}{\cancel{\sin x}} = \dfrac{1}{\sin x} = \text{cosec } x$ *	dM1 A1*
		(3)
(b)	Uses part (a) $\Rightarrow 7+\text{cosec } 2\theta = 3\cot^2 2\theta$	B1
	Either Uses $\pm 1 \pm \cot^2 2\theta = \pm\text{cosec}^2 2\theta \rightarrow$ 3TQ in cosec 2θ **Or alternatively** replaces cosec2θ with 1/sin2θ , cot$^2 2\theta$ with cos$^2 2\theta$/sin$^2 2\theta$, multiplies by sin$^2 2\theta$ and uses $\pm\cos^2 2\theta = \pm 1 \pm \sin^2 2\theta \rightarrow$ 3TQ in sin 2θ	M1
	$3\text{cosec}^2 2\theta - \text{cosec } 2\theta - 10 = 0$ **or** $10\sin^2 2\theta + \sin 2\theta - 3 = 0$	A1
	$(3\text{cosec } 2\theta + 5)(\text{cosec } 2\theta - 2) = 0$ **or** $(5\sin 2\theta + 3)(2\sin 2\theta - 1) = 0$ $\Rightarrow \text{cosec } 2\theta = -\dfrac{5}{3},\ 2$ **or** $\Rightarrow \sin 2\theta = -\dfrac{3}{5},\ \dfrac{1}{2}$ $\Rightarrow \sin 2\theta = -\dfrac{3}{5},\ \dfrac{1}{2} \Rightarrow \theta = \ldots$	dM1
	$\theta = \dfrac{\pi}{12}(0.262),\ \dfrac{5\pi}{12}(1.31),\ -0.322,\ -1.25$ (awrt these values)	A1, A1
		(6)
		Total 9

8(a)	States $\log a = 0.68$ or $\log b = 0.09$	M1
	$a = 4.79$ **or** $b = 1.23$	A1
	States $\log a = 0.68$ and $\log b = 0.09$	M1
	$a = 4.79$ **and** $b = 1.23$ **CSO**	A1
		(4)
(b)	The percentage of the population with access to the internet at the start of 2005	B1
		(1)
(c)	$P = 4.79 \times 1.23^{10} =$ awrt 38	M1, A1
		(2)
		Total 7

9(i)	$\int \frac{3x-2}{3x^2-4x+5}\,\mathrm{d}x = \frac{1}{2}\ln\left(3x^2-4x+5\right)(+c)$	M1, A1
		(2)
(ii)	$\int \frac{\mathrm{e}^{2x}}{\left(\mathrm{e}^{2x}-1\right)^3}\,\mathrm{d}x = -\frac{1}{4}\left(\mathrm{e}^{2x}-1\right)^{-2}(+c)$	M1, A1
		(2)
		Total 4

10(a)	$\dfrac{\mathrm{d}x}{\mathrm{d}y} = 12\sec^2 2y \tan 2y$	M1, A1
		(2)
(b)	$\dfrac{\mathrm{d}x}{\mathrm{d}y} = 12\left(\dfrac{x}{3}\right)\sqrt{\sec^2 2y - 1} \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}y} = 12\left(\dfrac{x}{3}\right)\sqrt{\dfrac{x}{3} - 1}$	M1, A1ft
	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\sqrt{3}}{4x\sqrt{x-3}}$	A1
		(3)
(c)	$y = \dfrac{\pi}{12} \Rightarrow x = 4$	B1
	$\dfrac{\mathrm{d}x}{\mathrm{d}y} = 12 \times \dfrac{4}{3} \times \dfrac{1}{\sqrt{3}}$ or $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{12\left(\dfrac{4}{3}\right)\sqrt{\dfrac{4}{3} - 1}}$	M1
	Correct $m_N = -\dfrac{16}{\sqrt{3}}$ o.e	A1
	$y - \dfrac{\pi}{12} = -\dfrac{16}{\sqrt{3}}(x - 4)$	dM1
	$y = -\dfrac{16\sqrt{3}}{3}x + \dfrac{64\sqrt{3}}{3} + \dfrac{\pi}{12}$	A1
		(5)
		Total 10